Fighting the Chamber

Our room will never pass a precision free-field standard at 250 Hz, and neither will most of the rooms in this guide. This is what twenty-five named facilities across fifteen countries actually do about it: the menu of methods, what each costs, the lowest frequency it works at, the error it leaves behind, and which of them we should use. Every number is quoted from a document we downloaded and read.

2026-09-11
campaign: chamber-landscape
7 threads · 35 new PDFs
evidence: papers/SoundVisualizer/chamber-landscape/EXTRACTS-*.md
our data: docs/arc-validation-remedies.pdf

00 The answer, before the evidence

IN SHORT Two different errors need two different cures, and only one of them is a room problem. The broadband error (±1.3 dB typical, 2.7 dB worst, present in every band) is ordinary reflection: fix it with geometry, then with a measured correction. The blade-tone error (up to 8 dB at 200–260 Hz) is either room modes or a propeller inflow bent by our own stand, and one afternoon with a loudspeaker decides which. Do not buy foam until that afternoon has happened.

1. **We are not unusual.** Small rotor chambers around the world cluster at a 63–275 Hz cut-off almost regardless of size, and small multirotor blade tones sit at 100–270 Hz. NASA Langley and TU Berlin both publish rotor measurements at or below their own facility's stated limit, and say so in print.
2. **Time gating, the method the world reaches for first, cannot help us in the band that matters.** Our shortest echo arrives 1.98 ms after the direct sound, so a gated response is trustworthy only above about 500 Hz. Reaching 250 Hz would need every reflector pushed out to 1.37 m of extra path.
3. **A measured room correction is the one method that fits our rig, and it has a published precedent with numbers.** A South African group corrected a fan facility that fails the sound-power standard outright, per position and per third-octave, to an uncertainty below 0.55 dB above 200 Hz. Their method is the loudspeaker session we had already planned for a different reason.
4. **Foam is the wrong lever at 250 Hz.** The quarter-wave rule is not negotiable: 34 cm of absorber for 250 Hz, confirmed independently by builds in Colombia and India to within a few millimetres. A few centimetres of foam is a 3 kHz treatment. The one exception found is a 100 mm metamaterial absorber measured at 99.2 % absorption at 239 Hz.
5. **The stand is a suspect, not a bystander, and someone has already run our experiment.** An Italian group using the same Tyto 1585 stand found its support pylon shifts directivity by up to 4 dB, and that reversing the direction of rotation is what reveals it.
6. **Report the split now.** Publish tone-notched broadband as the qualified product and blade tones as provisional, and state the frequency range we stand behind. The newer of the two qualification standards exists precisely so a room can qualify over whatever band it actually achieves.

01 Where we stand on the world's scale

IN SHORT Twenty-five named facilities in fifteen countries were read at source for this guide. Their low-frequency cut-offs cluster between 63 and 275 Hz whatever their size, because the cut-off is set by absorber depth, not by room volume. Small rotor chambers are genuinely small: Chiba University runs a hover stand in a room 3.86 × 1.56 × 2.4 m, NASA Langley's small hover chamber is 3.87 × 2.56 × 3.26 m. Our arc radius of 0.84 m sits inside the normal range too; Chiba uses 0.6 m. What separates us from them is not scale. It is that they have measured and published a qualification figure and we have not.

What the world's rooms actually achieve

FACILITY	COUNTRY	INTERNAL SIZE	CUT-OFF	WHAT THEY REPORT
NASA Langley, Small Hover Anechoic Chamber	USA	3.87×2.56×3.26 m	250 Hz	99 % absorption above 250 Hz; 8 mics, +44° to −49°, ≥10 rotor radii
NASA Langley, SALT	USA	337 m³	80 Hz	ISO 3745 deviations all in tolerance "except the farthest measurement locations below 250 Hz"
NASA Langley, LSAWT wind tunnel	USA	10.4 m cell, 0.61 m wedges	~200 Hz	revised to a "notional cut-on frequency of 175 Hz" from microphone-to-wedge distance
Amazon Prime Air hemi-anechoic room	USA	8.8×8.9×7.8 rotor diameters	—	"hemi-anechoic within ±1 dB above a blade passing frequency (BPF) harmonic of 2.2, but only within ±4 dB below that"
Virginia Tech (Eckel build)	USA	4.5×2.5×2.5 m	100 Hz	recirculation screens fitted downstream of each rotor
HKUST Aerodynamics and Acoustics Facility	Hong Kong	8.1×6.0×5.1 m	100 Hz	13 mics on a 2.5 m arc, 10° steps; drones fly free inside a safety cage
NWPU Unmanned System Research Institute	China	2.0×2.8×2.5 m	275 Hz	40-microphone ring at 1.47 m from the blades
Chiba University hover stand	Japan	3.86×1.56×2.4 m	200 Hz	4 mics on a 0.6 m arc at 0/25/50/75°; propeller blows upward, array downstream
CIRA semi-anechoic chamber	Italy	5.65×4.45×4.00 m	90 Hz	10 mics on an arc at 8 propeller radii; **same Tyto 1585 stand as ours**
Von Karman Institute, ALCOVES	Belgium	2.5×4.5 m main room	150 Hz	"free-field behavior down to 150 Hz for both broadband and tonal noise sources"
University of Bristol Aeroacoustic Facility	UK	7.9×5.0×4.6 m	160 Hz	23-microphone polar array at 1.75 m
TU Delft A-tunnel plenum	Netherlands	0.49 m wedges	173.5 Hz	passes ISO 3745 above 250 Hz at all distances; 125/160/200 Hz only to r = 1.5 m
PNRPU aeroacoustic chamber	Russia	0.8 m basalt wedges	125 Hz	"free acoustic field within radius of 2 m for tonal noise and 3 m for broadband noise"
ITMO, a radiowave chamber reused for acoustics	Russia	4.46×6.96×3.3 m	50–3150 Hz, 0.5–0.8 m only	"the largest deviation reaches the value of 15.5 dB"

FACILITY	COUNTRY	INTERNAL SIZE	CUT-OFF	WHAT THEY REPORT
University of Split	Croatia	2.8×1.7×2.05 m	250 Hz (traverse-limited)	passes both standards, yet "the measured range is small and not suitable for practical applications"
IIT (ISM) Dhanbad	India	2.75×2.40×2.98 m ext.	315 Hz measured	300 mm wedges; inverse-square law held "within 0.5 dB"
Universidad del Valle, Cali	Colombia	1.94×1.91×1.84 m	400 Hz nominal	about 200 USD per cubic metre; one point exceeds tolerance by 1.0 dB at 500 Hz
METU semi-anechoic room	Turkey	existing lab room	160 Hz	named failing bands: 63, 80, 100, 160 Hz depending on traverse
Korea Conformity Laboratories, Daejeon	South Korea	8.2×7.0×4.0 m	63 Hz	background below 15 dB(A); used for the indoor/outdoor drone comparison of §03
TU Berlin fully anechoic room	Germany	13.5×8.3×7.4 m	63 Hz	still reports "the BPF band of the UBADRON partly falls below the cut-off frequency"
Our arc, for comparison	Poland	room unrecorded; arc r = 0.84 m	**none measured**	tone error to 8 dB at 200–260 Hz; broadband residual ±1.3 dB, worst 2.7 dB

Sources, in order: Pettingill & Zawodny 2026 NASA/TM-20250003316 p.12; Rizzi et al. 2013 RASD p.10; Zawodny & Haskin 2017 AIAA pp.2, 16; Nardari et al. 2019 AIAA-2019-2497 p.2; Whelchel 2023 VT PhD p.57; Cao et al. 2025 Forum Acusticum p.1699; Xiao et al. 2022 Front. Mater. p.6; Sun et al. 2023 IJERPH p.3; Fasulo et al. 2025 Aerospace 12:647 pp.3–4; Gallo et al. 2025 Acta Acustica 9:16 p.2; Jawahar et al. 2025 Sci. Rep. 15:2170 p.3 and Hanson et al. 2023 JSV 559 p.3; Merino-Martínez et al. 2020 Appl. Acoust. 170 pp.5, 13; Palchikovskiy et al. 2016 AIP 1770:030116 pp.030116-1, -5; Bikmukhametov et al. 2026 arXiv pp.2, 5; Russo et al. 2018 Euronoise pp.2227–2228; Singh, Garg & Narayanan 2020 IJA 19 pp.57, 61; Orrego et al. 2018 Scientia et Technica 23(4) pp.471, 476–477; Kayhan 2008 METU MSc p.63; Kim et al. 2022 ICSV28 p.2; Hochbaum et al. 2026 QuietDrones pp.3, 9.

Specifics · what the comparison actually tells us

Cut-off is bought with depth, not with volume. The 13.5 m German room and the 8.2 m Korean room both publish 63 Hz; the 2.0 m Chinese room publishes 275 Hz. Two builds on different continents land on the quarter-wave rule within millimetres: Dhanbad's 275 mm wedge plus 25 mm base measures a 315 Hz cut-off against a 272 mm prediction, and Cali's 400 Hz design uses a 21 cm wedge against a 21.4 cm prediction. This is the number that governs our own options and it is arithmetic, not opinion.

Being small is normal; being unmeasured is not. Of the twenty-five facilities, the ones closest to our scale are Chiba (16 m³), NASA's hover chamber (32 m³), NWPU (14 m³) and Split (9.8 m³). Every one of them publishes a cut-off frequency and most publish a measured deviation. The distinguishing feature of our rig in this table is the empty cell.

Even the best-resourced laboratories publish below their own limit. TU Berlin, with a 63 Hz room, writes that "it should be noted that the BPF band of the UBADRON partly falls below the cut-off frequency of the anechoic chamber" (Hochbaum et al. 2026, p. 9). NASA Langley deliberately chose rotor speeds whose blade tones straddle its facility limit: "These rotation rates were selected for comparison since they correspond to BPFs that are near the anechoic cut-on frequency of LSAWT of approximately 200 Hz" (Zawodny & Haskin 2017, p. 16), and cautions that "caution should be used in quantitative comparisons at frequencies below the notional cut-on frequency of approximately 175 Hz". The honest practice is not to avoid the band. It is to name the limit and flag the data that crosses it.

What our own numbers look like against the standard tolerance. ISO 3745 allows ±1.5 dB up to 630 Hz, ±1.0 dB from 800 Hz to 5 kHz and ±1.5 dB above 6.3 kHz; six independent papers from six countries reproduce that table without variation. Applying those same limits to our position map, as an analogy rather than a conformity test, 29 of 209 cells fail on the tone-notched broadband map and 55 of 209 fail on the mixed-band map that the one-pager plots. The failures are not spread evenly: no cell fails at 0°, −18° or ±90°, while −72° through −36° and +18° through +72° account for all of them. That pattern is a geometry statement about our room, and it is the first thing the loudspeaker sweep should confirm or refute.

> **Read this as an analogy, not a qualification.** ISO 3745 qualifies a room by traversing a microphone away from a source and comparing the decay against the inverse-square law. Our map compares eleven fixed positions at one radius against their own mean. Same tolerance number, different test. We have never run the actual traverse, which is why the cut-off column above is empty for us and will stay empty until we do.

02 Why the wall exists, and why it sits exactly on our blade tone

> IN SHORT An absorber only works once it is about a quarter of a wavelength deep, which is 34 cm at 250 Hz. Below that frequency a small room stops behaving like open air and starts behaving like a resonator, and a pure tone and a band of noise stop agreeing with each other. That last point is the one that matters most for us, and it is measured, not theoretical: in the same chamber, on the same traverses, a pure tone failed the free-field tolerance in 29 % of samples at 250 Hz where broadband noise failed in 4 %.

Specifics · the three mechanisms, with their numbers

1. The quarter-wave tax. "The most widely accepted approximate theoretical formula to determine the cut-off frequency of the chamber is f = c/4L" (Singh, Garg & Narayanan 2020, p. 58), where L is the wedge length. The materials literature states the same rule from the other side: "the sound absorption peak appears at a frequency where the thickness of the porous material corresponds to one-fourth of the wavelength of the sound wave" (Sakamoto et al. 2023, *Materials* 16:5671, p. 16). At 250 Hz that is 34 cm. A 2 cm foam layer has its own quarter-wave frequency at about 3.4 kHz; a 10 cm layer at about 860 Hz.

Measured thin samples confirm it is not a soft rule. Felted mountain-sheep wool at 9.9 and 19.5 mm: "For both samples, at low sound wave frequency the α coefficient was low, from 0.1 to 0.15. Above 500 Hz, when the frequency grew, the α coefficient increased significantly" (Kobiela-Mendrek et al. 2022, *Materials* 15:3139, p. 8). A 5 mm rice-husk and bagasse panel reaches a low-frequency coefficient of 0.146 to 0.225 (Bidin & Mohd Sari 2025, p. 123). Bare coconut coir at 10 to 20 mm does not reach its own absorption peak until 3.7 to 5 kHz (Zulkifli et al. 2010, p. 263).

And the wedge specification is not the same thing as the room's behaviour. The Russian wedges of §01 are specified so that the "absorption coefficient at normal incidence of sound is αn ≥ 0.99 at frequencies f ≥ 100 Hz", yet "Certification of the AC2 AC according to ISO 3745:2003 showed that this chamber corresponds to the requirements imposed on ACs for a frequency range of f > 200 Hz" (Belyaev et al. 2015, pp. 608–609). A full simulation of an optimised 100 Hz wedge found it met the free-field tolerance over only 57 to 66 % of the test line at its own design frequency, because "the 99% energy absorption means 10% pressure reflection, which is still very large" (Jiang, Zhang & Huang 2016, *JSV* 381, pp. 152–153).

THE MODAL BOUNDARY

2. Below the Schroeder frequency a room is a set of resonances, not a free field. The frequency is $f_s = 2000\sqrt{T/V}$. A round-robin paper tabulates it for a small room of 145 m³ with a 2.0 s reverberation time and gets **234 Hz** (Brinkmann et al. 2019, *JASA* 145(4), Table II, p. 2747) — inside our own 216 to 258 Hz tone cluster, in someone else's room. For a room of the order we believe ours to be, 3 × 3 × 2.5 m with thin foam, the same formula lands between 190 and 270 Hz depending on the reverberation time. Our blade tones sit on that boundary. We cannot compute it properly until the room is measured, which is the first item of §07.

Why the boundary matters: above it the room transfer function is statistical, and "its frequency average (in 1/3 octave bands or critical loudness bands) is robust against those changes", while the fine structure of magnitude and phase is not (Vorländer 2013, *JASA* 133(3), p. 1204). A third-octave band of noise averages over that fine structure. A single blade tone is one sample of it.

TONES AND BROADBAND DISAGREE, MEASURED

3. This is the best-evidenced statement in the whole guide. Georgia Tech measured the same chamber, on the same traverses, with a pure tone and with broadband noise:

FREQUENCY	PURE TONE, SAMPLES OUTSIDE TOLERANCE	BROADBAND NOISE, SAMPLES OUTSIDE TOLERANCE
80 Hz	56 %	54 %
125 Hz	53 %	30 %
250 Hz	29 %	4 %
500 Hz	37 %	0 %
1000 Hz	16 %	0 %

Cunefare et al. 2003, *JASA* 113(2), Table IX, p. 889, "fixed reference" columns for one of the two traverses.

> **"In contrast to the results obtained with the pure tone excitations… the deviations observed using broadband noise are unremarkable… the use of pure tones is a much more stringent test of the chamber's performance than the use of random noise."**
> Cunefare, Badertscher, Wittstock & Biesel, JASA 113(2), pp. 889–890

Three independent confirmations. A professionally built 51 m³ chamber "could satisfy the ISO tolerances when using random noise but failed to qualify when using pure tones", with a fitted 2056 Hz tone decaying at "eight decibels per doubling — a value that is physically impossible" (Nash 2019, *ICA*, pp. 1343, 1348). Perm publishes two different working radii for one room: "a free acoustic field within radius of 2 m for tonal noise and 3 m for broadband noise" (Palchikovskiy et al. 2016, p. 030116-1). And on a real drone rather than a loudspeaker, adding one reflecting floor moved the broadband overall level by about 3 dB but the blade tone by 10 dB: "At the BPF, which is the dominant tonal contribution, the difference in the measured SPL can be as significant as 10 dB" (Ma et al. 2022, Quiet Drones, p. 7).

> **What this licenses us to say.** Our own result — tones bent up to 8 dB while the broadband beside them stays within ±1.3 dB — is not an anomaly and not a sign of a broken measurement. It is the standard behaviour of a room near its modal boundary, reported by at least four independent groups, one of them on the same kind of source. It also means the two halves of our data deserve different treatment and different confidence, which is what §03 and §06 are about.

03 The methods menu

> IN SHORT Ten things can be done about a room that is not good enough, and they are not alternatives to each other: they are a sequence. Move the geometry first because it is free. Report the split second because it is free. Measure a correction third because our rig is the one kind of rig a correction actually works on. Treat the surfaces fourth, only where the sweep names them. Everything else is either unavailable to us at 250 Hz or is a research project.

METHOD	LOWEST USABLE FREQUENCY	RESIDUAL ERROR	COST	FAILS WHEN	FOR US
Move the geometry	sets everything else	–	a tape measure	nothing can move	**first**
Report the split, state the range	n/a	removes no error, hides none	free	a reader needs one number	**now**
Measured room correction	200 Hz (published)	<0.55 dB above 200 Hz	1 speaker + 1 reference run	anything moves afterwards; SNR too low	**yes**
Average over source speed	no floor	unquantified in literature	one longer session	tone pattern is speed-independent	**yes**
Time gating	**500 Hz here**	1–2 dB where curves join	software only	first echo arrives too early	**above 500 Hz**
Absorb it (wedges)	34 cm of depth per 250 Hz	design vs effective cut-off differ 20–28 %	~200 USD/m³, ×4 per octave down	no room for the depth	**not at 250 Hz**
Subwavelength absorber	185 Hz at 100 mm	99.2 % absorption at 239 Hz	fabrication, not purchase	narrowband; must be built	**the one exception**
Active absorption	to 430–600 Hz	9.6 dB average field reduction	a PhD	—	**research**
Change venue: outdoors	background-limited	0.1 dB vs chamber at 20 dB SNR	weather station, no chamber	SNR below ~15 dB; wind	**for absolute levels**
Change instrument: intensity, holography, arrays	50 Hz (intensity)	1–3 dB (grade 1–2)	probe or array we do not own	strongly reactive fields	**not now**

1 · Move the geometry. It sets the ceiling on every other method

Every other method's lowest usable frequency is decided by how far the nearest reflector is. Our measured echoes are an extra path of 0.68 m at −72° and 1.49 m at +36° and +54°, which is 1.98 and 4.34 ms. The gate length available to us is therefore 1.98 ms and nothing can change that except moving something.

TO WORK DOWN TO	EVERY REFLECTOR NEEDS AN EXTRA PATH OF	WE CURRENTLY HAVE
1000 Hz	0.34 m	0.68 m worst case
500 Hz	0.69 m	
250 Hz	1.37 m	
125 Hz	2.74 m	

The rule behind the table is quoted directly: "It is recommended that the gate length should be 5ms or more (dictated with a time-bandwidth requirement for correct frequency response estimation on frequencies above 200Hz). Practically this means that for a correct measurement setup we must place the microphone at the distance h from the nearest wall or floor so that time delay of first reflection would be larger than 5ms" (Matelján, ARTA Application Note 4, p. 11). His worked example needs a metre of clearance to reach 200 Hz at a 0.48 m source distance.

The rig literature adds its own clearance numbers, which are about the source rather than the room. A generic airframe element 0.1 to 0.2 rotor radii below the blade tip changed the level by 8 to 10 dB, and "Increasing this distance to 0.4 tip radii yields… levels… nearly identical to those for the case of an isolated rotor" (Zawodny & Boyd 2017, p. 6). Ground effect is negligible by four rotor radii and strongest at 0.75 (Jawahar et al. 2025, p. 6).

> VERDICT **Do this first, because it is free and because it sets what methods 3 and 5 can achieve. The action is to record the room, find the 0.68 m surface, and move it or move the arc.**

2 · Report the split and state the qualified range

The newer qualification standard exists precisely to let a room claim the band it has rather than the band it wishes for. Under ISO 3745 a room is in full conformity only if it holds tolerance across 100 Hz to 10 kHz; "ISO 26101 does not specify any required frequency range and allows a free-field environment qualification provided that the one-third-octave bands comprising the frequency range are contiguous" (Russo et al. 2018, pp. 2227–2228). Facilities use that licence openly: an Italian group integrates its overall levels only above its own cut-off, with "the integration bounds f1 and f2… chosen as 90 Hz (cut-off frequency of the chamber) and 20 kHz… thus confining the analysis to the audible range and excluding room-reflected energy below 90 Hz" (Fasulo et al. 2025, p. 7). A Russian chamber publishes a table of frequency-and-distance cells in which it is usable and stays inside them.

Where a chamber fails, the standard compels disclosure rather than silence: if background-noise criteria are not met "the report shall clearly state that the background noise requirements of this International Standard have not been met… Furthermore, the report shall not state or imply that the measurements have been made 'in full conformity' with ISO 3745" (ISO 3745:2012, §5.2.3, p. 9).

> VERDICT **Free, immediate, and already half-built: the tone and broadband split now exists in our own software. What is missing is the sentence that says which band we stand behind.**

3 · Measure the room's transfer function and subtract it

This is the method that fits our rig, and the case for it is a single paper from Stellenbosch that corrected a fan facility which does not conform to the sound-power standard at all. Their trigger is worth quoting because it is our situation exactly: "measurements in accordance with this standard [ISO 3744] are only valid where the environmental/reflection correction in decibels is smaller than or equal to 4 dB, which is rarely the case for fan test facilities of which the purpose is performance testing" (du Plessis, Van der Spuy & Reuter 2022, p. 24).

The method is a small loudspeaker measured once in a real anechoic chamber and again at the same positions in the working facility, third-octave by third-octave, and the difference taken. The result: "On average, for the three microphone positions, reflections cause a 5.65 dB increase at a frequency of 1000 Hz, and even more at lower frequencies… it is 1.52 dB at a frequency of 5000 Hz", and after correction "the uncertainty in measurements is minimal (below 0.55 dB for frequencies above 200 Hz)" (p. 29). They also show the generic correction is not a substitute: following the blanket 6 dB cap of a general noise standard "underestimates the effect of sound reflection in the very low frequency range… Therefore, each test facility in which fan noise measurements are taken should be characterised for sound reflections at each 1/3-octave frequency, independently" (p. 29).

> **The tension you must know about, because our own remedies report leans the other way.** A drone-measurement paper states flatly that "In practice it may be possible to correct the data for the comb-filtering effect as long as the source signal is broadband noise. However, if the source signal contains pure tone components, it will not be possible to do this correction" (Rasmussen & Winberg 2022, p. 4). Both statements are true and they are not about the same situation. Rasmussen and Winberg are describing a drone at an arbitrary, varying height above a ground plane, where the comb pattern moves with the geometry. Du Plessis corrected a fixed source at fixed microphone positions and validated the result per band. **Our arc is the fixed case:** eleven microphones bolted at known angles, a source that returns to the same hub. That is what makes the correction admissible here and inadmissible for a flying drone.

The limit on that admissibility is measured, and it is tight. Moving a loudspeaker by one centimetre and reusing the same compensation filter "causes amplitude variation of approximately ±6 dB in the simulated free field response" at higher frequencies (Bellmann & Klippel, p. 7). A correction is locked to the geometry it was measured in. If a microphone is unclamped and re-seated, the correction for that position is void.

> **And a caution that lands exactly on our frequency.** A Swedish thesis measured the same chamber and the same microphone positions twice, once with a calibrated loudspeaker and once with the real engine under test, and found "there is correlation between the results from the loudspeaker and engine but not in every frequency bands especially not in the regions of 200 and 250 Hz. This could be the effect of reflection or radiation from the test cell floor, since there are some metal sheets there, which can be excited by engine vibrations via mountings" (Mehrgou 2012, KTH TRITA-AVE 2012:42, p. 34). A substitute source is not guaranteed to reproduce what the real source does, and the band where it failed for them is the band we care about. The lesson is not to abandon the method but to test it: our loudspeaker session must include a comparison against the propeller at the same positions, and if they disagree at 238 Hz that disagreement is itself the finding.

VERDICT **The strongest single option we have. It needs the loudspeaker session already planned in the remedies report, plus one reference run in a real chamber, which is the second reason to ask the university for chamber time.**

4 · Average the tone away across source speed

Our own data is the argument here: the tone polar is round at 216 Hz, has an 8.3 dB hole at −36° at 238 Hz, and a 7.8 dB hole at −72° at 258 Hz, while the broadband floor beside it barely moves. A pattern that changes completely over a 19 % change in frequency is a pattern that averages out over a speed ladder. The standards side sanctions the underlying idea, permitting either signal type for qualification but requiring finer spatial sampling for tones, "for discrete-frequency measurements using pure tone signals, the microphone may be moved slowly and continuously along the traverse" (ISO 26101-1:2021, §5.1.4.3, p. 4). Cunefare's own conclusion is that a tonal qualification needs sample spacing of "15% of a wavelength at the frequency of interest" (2003, p. 892), which is the spatial analogue of the same idea.

VERDICT **Cheap, uses hardware we already have, and already scheduled as the speed-ladder session. No published paper quantifies the residual after such an average, so we would be producing that number ourselves.**

5 · Time gating, and why it stops at 500 Hz in this room

The technique separates the direct sound from the reflections by discarding everything after the first echo. The cost is bandwidth: the usable lower frequency is roughly the reciprocal of the gate, demonstrated in the source as "the time-bandwidth requirement is satisfied on frequencies above 177.9Hz. (Note: 1/gate = 1000/5.604 = 178.4)" (Matelján, p. 8). The failure below that limit is not noise but a wrong answer, because the window no longer contains a full period.

An industrial variant replaces the single gate with a scan on two surfaces and a holographic separation, reaching below 100 Hz, at the price of 100 to 200 measurement points and 15 to 30 minutes per source (Bellmann & Klippel, p. 5). A road-surface standard broadens the window by subtracting a separately measured free-field signal instead of gating it away, noting that this "allows broadening of the time window, leading to a lower frequency limit of the working frequency range, without having very long distances between loudspeaker, microphone and surface under test" (ISO 13472-1, Clause 4.2).

VERDICT **Useful to us for naming the reflectors and for everything above 500 Hz. It cannot answer the 238 Hz question, and no amount of processing changes that: the limit is set by the room's geometry, not by the software.**

6 · Absorb it, and what that really costs at 250 Hz

A conventional wedge answer to our band is 34 cm of depth on every surface that matters. The cost scaling is published: a Colombian build came to "approximately 200 USD per square meter of effective built volume" for a 400 Hz chamber and "If a lower cutoff frequency is intended, e.g 100 Hz, the cost per square meter would increase in approximately four times" (Orrego et al. 2018, p. 477). Hybrid wedges buy some of it back: "If a further reduction in cut-off frequency is required without changing the space of an anechoic chamber, the hybrid wedge is a good choice" (Wang & Tang 1996, pp. 109–110), and a flat-walled graded profile matched a classic wedge's 100 Hz target in 0.762 m rather than 0.800 m (Jiang et al. 2016, Table 3).

VERDICT **Not the lever for 200–260 Hz in a room our size. It is the right lever for the 630–1000 Hz reflectors, where ten centimetres of ordinary foam on the named surface is enough.**

7 · The one treatment that reaches our band in a hand-sized depth

A coiled-channel metasurface absorber, tested to ASTM E1050-12, reaches "99.2% absorptance at 239 Hz in experiment" in a total thickness of 100 mm, and as a broadband device shows "high absorptivity (>80%) exceeding one octave from 185 Hz to 385 Hz" (Long et al. 2020, *Sci. Rep.* 10:13823, pp. 1–3). A plain quarter-wave absorber at 236 Hz would need 363 mm. This is the only treatment found anywhere in the campaign whose measured performance lands on our exact frequency at a depth our room could accept.

VERDICT **Worth keeping on the table, but only after the loudspeaker session. If the 238 Hz pattern turns out to be inflow rather than the room, this absorber would treat a problem we do not have.**

8 · Active absorption

Two routes exist. Driving a secondary source inside a perforated-panel absorber's backing cavity pushed its useful ceiling from 215 Hz to 430 Hz with absorption "close to 1 after control" (Ma et al. 2022, *Appl. Acoust.* 185, p. 11). Cancelling the whole reflected field, reconstructed from a sensor ring, gave an average 9.6 dB reduction up to about 600 Hz and cut reverberation times by a third to a half (Haasjes 2025, Twente PhD, pp. 112, 118), motivated explicitly because "a typical acoustic anechoic chamber has a lower cut-off frequency of up to 200 Hz... Active noise control is effective at lower frequencies".

VERDICT **Real, current, and a doctorate's worth of work. Noted so that we know the ceiling of what is possible, not proposed.**

9 · Change the venue

Outdoors. The cleanest number in the whole campaign: the same drone measured indoors and out gave "82.6 dB and 82.5 dB, respectively. Therefore, it may be possible to measure drone noise as much as a laboratory even outdoors if only the difference from the background noise can be sufficiently secured" (Kim et al. 2022, p. 6). The condition is explicit in the same paper: the octocopter cleared background by more than 20 dB and agreed; the quieter quadcopter cleared it by only 10 dB and did not. Regulators accept outdoors as the normal venue and manage repeatability with wind limits, requiring average wind at or below 2.6 m/s for hover (EASA 2023, §Noise.UAS.008).

Half a chamber. Removing the floor treatment costs about 3 dB on broadband overall level but up to 10 dB at the blade tone, measured on real drones with a repeatability of 0.5 dB, so the difference is real (Ma et al. 2022, p. 7). A ground board turns the reflection into a constant instead of a comb, at the price of a fixed 3 or 6 dB offset, but a real finite board scatters at its edges and "the corrections cannot be directly applied to the experimental data" (Whelchel 2023, p. 67).

Borrow a real one. The Politechnika Warszawska electroacoustics department lists a 300 m³ anechoic chamber, B&K laboratory sound sources and impulse-response software, and offers measurements to outside parties.

VERDICT **Outdoors is a genuine route to absolute levels for a loud enough rotor and is worth one comparison session. Borrowing the university chamber is the cheapest way to get the reference run that method 3 needs.**

10 · Change the instrument

Sound intensity is a vector and survives a reverberant field that would corrupt a pressure measurement; the standard says so itself, warning that a pressure level "may contain strong contributions from sources outside the measurement surface, but may be associated with little net sound energy flow, as in a reverberant field in an enclosure" (ISO 9614-1:1993, introductory note). Its accuracy grades give 1 to 3 dB. Near-field holography that models the boundary rather than merely subtracting it took a greater than 14 dB contamination down to under 1 dB in a controlled validation (Lin & Li 2021, *Sensors* 21:5521, pp. 8–9). Phased arrays are the disappointment: adding an image-source model to a deconvolution in a hard-walled tunnel cleaned up the maps but "has little appreciable effect on the spectral levels", and an 8 dB facility-to-facility gap survived it (Bahr 2021, NASA, p. 11).

VERDICT **None of these is available to us this year. The array result is worth remembering as a caution: a method can fix the picture without fixing the number, and the number is what a directivity plot is made of.**

04 Qualification: how to know, and what to say when you fail

IN SHORT The test is simple and we have never run it: move a microphone away from a source and check that the level falls by 6 dB per doubling of distance, within ±1.5 dB below 630 Hz and ±1 dB from 800 Hz to 5 kHz. Two standards define it and practitioners criticise both. What nobody does is stay silent: every facility in this survey that found a shortfall restricted its radius, restricted its frequency range, or printed the caveat.

Specifics · the tolerance, the two standards, and the honest disclosures

The tolerance table is the one thing in this literature that never drifts. Six write-ups from the United States, Croatia, the Netherlands, Russia twice and India reproduce it identically: ±1.5 dB at or below 630 Hz, ±1.0 dB from 800 to 5000 Hz, ±1.5 dB at and above 6300 Hz. The UAS standard published in 2024 carries the same numbers, starting its table at 125 Hz.

The two procedures differ in geometry, not in tolerance. ISO 26101 tightened the source-stability requirement from ±0.5 to ±0.2 dB, specified the traverse directions precisely rather than loosely, and required spacing no coarser than a tenth of a wavelength below 1 kHz. It also decoupled qualification from the sound-power use case, which a chamber manufacturer describes as making qualification "more readily available for audio, automotive, military, and other users that are not interested in sound power measurements" (Winker & Stahnke 2016, p. 1).

Both are criticised, from opposite directions, by people who build and use chambers. On the older one: "By using this fitting method, a chamber can fully comply with ISO 3745 at certain frequencies while not exhibiting free-field performance at those frequencies" (Winker & Stahnke, pp. 2–3) — that is, the curve fit can absorb a real error. A Colombian group reached the same verdict independently a decade earlier, concluding that applying ISO 3745 "es insuficiente para la calificación de cámaras anecoicas" (Quintana Soler et al. 2008, p. 10), and noted that even their outdoor, genuinely free-field reference measurement did not cleanly pass it. On the newer one: its traverse-length rule "will require future chambers to unnecessarily grow larger while reducing the frequency range of existing chambers based on geometry and not acoustical performance" (Winker & Stahnke, p. 5). Split's chamber is the illustration: its 250 Hz floor is set by how far the microphone can travel, not by the wedges.

How much the room contributes to the final answer, in an uncertainty budget. The National Physical Laboratory measured it and found it can dominate: room-alone worst-case deviations "just over 3 dB for the ISO 3745:1977 qualification and reaching just over 11 dB for the current ISO 3745:2003 procedure", with A-weighted sound-power standard uncertainties of "0.5 dB, 0.8 dB and 0.8 dB" under the old procedure against "3.7 dB, 4.2 dB and 4.2 dB" under the newer one, concluding that "the current room qualification procedure as specified in ISO 3745:2003 is inadequate and significant errors in sound power level determination may occur" (Simmons, Jobling & Payne 2004, NPL DQL-AC 007, p. 45).

A calibration for how good acoustics can be when the room is not the problem. Six national metrology institutes on three continents calibrated the same travelling microphones in their own free-field facilities. Every pairwise difference at 2.5 kHz was inside ±0.15 dB, and the worst at 20 kHz was just under 0.2 dB (Barrera-Figueroa et al. 2010, CCAUV.A-K4, Tables 12 and 15). That is two orders of magnitude tighter than the ±1 to ±1.5 dB a chamber is allowed to deviate. The loose tolerance is not a statement about what acoustics can achieve. It is a statement about how hard large-room free-field qualification is.

What labs do when they fail. Restrict the frequency range, which the standard itself provides for. Restrict the radius, as the Russian radiowave chamber does with a published table of usable frequency-and-distance cells. Name the cause and stop, as in "the chamber is absorptive but not anechoic" (Nash 2019, p. 1349). Accept a stated shortfall, as in an array radius that "fell slightly short of the 2L condition… with a 13% deviation deemed acceptable" (Cao et al. 2025, p. 1699). Or move the offending object, since "if any object is disturbing the free field acoustic region, the object can be shifted to different safe location" (Rajmane & Baumann 2016, p. 333). None of the sources read for this campaign says nothing.

05 The rig, not the room

IN SHORT Half of our 8 dB problem may not be the room at all. Fifty years of static-fan testing shows a support structure near a rotor makes the inflow lopsided and the blade tone one-sided, by 6 to 10 dB, on the structure's own side. An Italian group running the same Tyto 1585 stand we use measured its pylon shifting directivity by up to 4 dB, and found that reversing the direction of rotation is what exposes it. That is a one-afternoon test we can run before spending anything.

Specifics · what a stand does to a tone

The classical result is from a NASA fan rig whose support sat behind the inlet lip: "Atmospheric turbulence effects are considered to act more or less uniformly around the circumference whereas rig interference, which is roughly an equal contributor to the distortion, acts mainly at the bottom of the inlet" (Hanson 1977, NASA CR-2899, p. 1), with distortion "2 to 3 times stronger at the bottom than at the top" (p. 24) and the consequence stated plainly: "The localized blockage causes a noise directivity pattern which is not symmetric with respect to the fan axis of rotation" (p. 21). A single probe left in an inlet raised the tone by "at least 6 dB… on the axis as well as 7 dB at 30°", fading to nothing by 90° (Hodder 1977, p. 7).

Screens fix the turbulence, not the structure. An inlet flow-control device bought "about a 5 dB reduction in blade passing tone" and up to 10 dB at some angles (Woodward et al. 1978, p. 2), but a lobe caused by the rotor's own wake striking a fixed structure downstream "is essentially the same with and without the ICD installed" (Homyak et al. 1983, p. 6). Removing the broadband turbulence can even reveal a lobed pattern that was previously masked (Woodward & Glaser 1980).

The single most useful paper found in this campaign. An Italian aerospace research centre ran an isolated rotor on a **Tyto Robotics 1585 thrust stand**, the same model as ours. They report "a small but significant mounting asymmetry: although the rotor hub is centred, the pylon supporting the isolated propeller is offset to the left side of the array… That lateral offset produces additional shading behind the propeller and contributes to the left-right level differences observed in the OASPL profiles." Reversing the sense of rotation "reveal[ed] far larger azimuthal variations than those observed when only the blade was changed", reaching "about 3 dB at 6000 rpm and 3–4 dB in the 7000–8000 rpm range", and their conclusion is that "even modest geometric asymmetries, here the support pylon, can shift directivity by up to 4 dB" (Fasulo et al. 2025, pp. 10–11, 24). The reversed-rotation run already in our campaign plan is therefore not a speculative idea. It is a replication.

Recirculation is a different, well-mapped hazard. In a chamber of about 32 m³ the flow turns over within seconds and inflates the harmonics: "flow recirculation results in a significant increase in higher harmonic noise, with an increase of more than 15 dB in some harmonics", while "the first harmonic of the blade passage frequency is relatively unaffected" (Stephenson, Weitsman & Zawodny 2019, pp. 1153–1154). The clean window was "sustained from 3.5 to 7.4 s" and downstream meshes extended it from 3.9 to about 10 s (Weitsman et al. 2020, pp. 1327, 1335). In a chamber 8 to 12 times larger the same mechanism took about 30 s and barely touched the tones. Our captures are 2 s per step, which is inside every clean window reported; the effect is also largest out of the rotor plane, and our arc sits in it.

A blind spot in the field, which our rig happens to be pointed at. The recirculation papers all measure along a single azimuth and say why: "An azimuthal array was not employed as the acoustic emissions of an isolated rotor in hover should be symmetric" (Stephenson et al. 2019, p. 1153). A confinement study "assuming an axisymmetric acoustic field" in its analysis method (Nardari et al. 2019, p. 9). The assumption is reasonable and almost never tested. Our flat-arc configuration tests it directly, which is why our result looks strange against this literature: most of the literature could not have seen it.

And the scatter this class of rig normally achieves. Run-to-run and microphone-to-microphone agreement is reported at 0.5 to 0.6 dB by three independent groups, with the same Italian stand giving "changes in blade manufacturing tolerance altered azimuthal OASPL by less than 0.6 dB across 3000–8000 rpm" (Fasulo et al. 2025, p. 24) and a NASA spectral-processing floor of ±0.6 dB. Our broadband residual of 1 to 3 dB is two to five times that, which is independent support for our own finding that 99 of 209 cells sit more than three standard errors from a circle. It is a real effect, not measurement noise.

06 What the world requires, and where it is heading

> IN SHORT There is no drone noise limit anywhere in the world yet. Europe and the United States both published measurement methods with the limits clause left empty, and both measure a single microphone on the ground under a flying aircraft. The one standard that treats directivity as a first-class goal is ISO 5305, published in 2024, and it exists because the regulation before it ignored directivity. That is the standard our arc is aligned with, and it lets the user choose between a chamber, a wind tunnel and the open air.

Specifics · four documents and one gap

ISO 5305:2024 covers rotor-powered unmanned aircraft below 150 kg and states in its own scope that "It does not account for the UAS noise certification or regulation". It specifies three test facilities, chamber, anechoic wind tunnel and outdoors, and says "the selection of the facility is not compulsorily specified" (§6.1). It asks for 20 Hz to 20 kHz at 48 kHz sampling or better, a microphone distance of at least five vehicle diameters, and a background margin "at least 6 dB, preferably more than 15 dB" below the source. For the chamber it delegates qualification to ISO 26101-1 and carries the familiar tolerance table starting at 125 Hz.

Its Introduction says why it exists, and it reads like a description of our instrument's purpose: the earlier European rule "measures the sound power level of UAS in indoor and outdoor environments… However, the measurement is limited only to the UAS in hover. In addition, the microphone arrangement is not specified to avoid the influence of unsteady aerodynamic flow induced by the UAS on the acoustic measurement. Also, the directional nature of the noise produced by UAS was not considered." Against that, "Configurations of the microphones are specified to ensure measurements are made at different locations to quantify the directivity of UAS noise."

EASA's guidelines for aircraft under 600 kg are voluntary, outdoor, and single-microphone: an inverted pressure microphone 7 mm above a 40 cm ground plate, a level flypast between 17 and 150 m and a 30-second hover, at least six runs, wind at or below 5.1 m/s for the flypast and 2.6 m/s for hover, and a signal at least 15 dB(A) above background. The document's noise-limits section reads, in full, "SUBPART D – NOISE LIMITS (RESERVED)".

The FAA has no general rule either and certifies model by model: it is "making available noise certification standards that apply to individual model unmanned aircraft… because no generally applicable noise certification standards were available". Six aircraft have been handled this way. Its first such rule put the microphone on a pole rather than a ground board because "a pole-placed microphone is relatively simple and less costly to deploy", and it used few microphones deliberately: "The proposed rule was designed to be simpler for UA by using fewer microphones."

ICAO is drafting, not publishing: work has begun on "harmonized noise certification practices" for drones and larger air-mobility aircraft, to appear as "an ICAO circular on 'Interim noise measurement guidelines for smaller emerging technology aircraft' in 2025", with "unresolved topics such as noise limits or applicability weight range" deferred to the next cycle. Brazil's and Japan's authorities have "shared with WG1 similar intentions".

The gap, elsewhere. China's mandatory drone safety standard GB 42590-2023 contains a clause 4.14 titled "Noise", whose text is paywalled and which we did not read. India's Drone Rules 2021 contain the word noise zero times across all thirty-seven rules. A Korean review reports that noise evaluation items are absent both domestically and, in its authors' assessment, internationally, and that in the absence of a standard procedure each manufacturer and research institute uses its own method, which makes objective comparison between aircraft difficult (Cho et al. 2024, p. 528; translated from the Korean by the research thread, so paraphrased here rather than quoted).

> **Where our arc sits.** Nobody's standard specifies an array like ours, because directivity arrays are being invented ahead of the standards: a hemispherical 18-microphone array in Belgium, a 10-microphone arc in Italy, a 64-microphone distributed array in Germany, 13 microphones on a 2.5 m arc in Hong Kong. Our eleven-microphone arc is a member of that family rather than of the regulatory single-microphone family. The corollary is that we are not measuring for a certificate, so the question "does our room pass ISO 3745" is the wrong question. The right one is "how large is our uncertainty, and do we state it", which is what §04 and §07 are for.

One facility in this guide is not a chamber at all, and it is the closest to our situation. The Stellenbosch fan rig is a performance-test facility retrofitted for acoustics, whose authors say outright that purpose-built performance rigs routinely fail the qualification threshold. It is the only entry here that solved the problem by measuring and correcting its own room rather than by building a better one, and its corrected uncertainty, below 0.55 dB above 200 Hz, converges on the same 0.5 to 0.6 dB that small-rotor rigs in Italy, Hong Kong and the United States report as their repeatability floor — across a completely different rotor scale, facility type and continent.

One more current from the same literature: psychoacoustic metrics are already routine in drone work, with loudness to ISO 532-1, tonality from ECMA-74 or ECMA-418-2 and sharpness to DIN 45692 all in use alongside the emerging standard. Our tool already computes loudness and roughness, which puts it on the right side of that trend.

07 What we should actually do, in order

> IN SHORT Six steps. The first four cost nothing but time and they decide whether the rest are needed at all. Nothing here requires buying absorber, and nothing here should be bought before step three has told us what we are treating.

STEP	WHAT IT IS	COST	WHAT IT BUYS
1. Record the room	Hub height, room dimensions, foam type and thickness, what stands within 1.5 m of the arc ends, where the beam sits	an hour	the Schroeder frequency, the plane fit, and the ability to predict which surface is the 0.68 m reflector
2. Reverse the rotation	One capture set with the propeller turning the other way, everything else identical	an afternoon	separates stand-induced asymmetry from room asymmetry, replicating the Italian result on the same stand model
3. Loudspeaker session	Sweep from the hub, all 11 mics, motor off; then tones at 216, 238 and 258 Hz; then the box rotated 45°	an afternoon plus the sweep tooling	names the two reflectors, decides room versus inflow, and produces the per-position correction of method 3 — and must be cross-checked against the propeller at the same positions, per the Swedish caution in §03
4. Speed ladder	12–15 steps from 1750 to 2050, blade frequency read from the audio, microphones rotated one place per run	a day	averages the tone error and separates microphone from position without relying on the two historical swaps
5. Reference run in a real chamber	The same loudspeaker, the same positions, in the university's 300 m³ chamber	one visit	turns the room measurement into an absolute correction instead of a relative one

STEP	WHAT IT IS	COST	WHAT IT BUYS
6. Treat or move what was named	Ten centimetres of foam on the 630–1000 Hz surface, or move the arc away from it	`small`	the only treatment step, and only for surfaces the sweep actually identified

The one piece of software that gates all of this. There is no sweep generator, no playback path and no deconvolution script in the repository. Steps 3 and 5 cannot happen until that exists. It is a few hours of work and it is the next thing to build.

What we will be able to claim afterwards. If step 2 shows the pattern follows the rotation, the 238 Hz error belongs to the stand and no room treatment will touch it; the cure is to move the stand out of the inflow or to report tones as stand-affected. If step 3 reproduces the 8 dB pattern from a loudspeaker, it belongs to the room, and methods 3 and 4 apply. Either way we finish with a stated qualified range, a measured correction for eleven fixed positions, and an uncertainty figure — which is what every facility in §01 publishes and we currently do not.

08 What we still do not know, and what to fetch

> IN SHORT Four honest gaps in this guide, and one paper whose absence matters more than the rest: a 2025 study of a drone in a reverberant room corrected with a tailored Green's function. It is the only paper found anywhere that does exactly what §03 method 3 proposes, on exactly our kind of source. It is open-access and still could not be fetched.

Gaps in the evidence, stated plainly

- **No chamber-versus-chamber round robin exists** in anything we could read. The nearest equivalent is a microphone-calibration key comparison between national laboratories, which is a different measurement. So the honest answer to "how much do two good chambers disagree on the same rotor" is that we could not find one; the closest is a within-NASA comparison of two facilities reported qualitatively as "excellent" agreement in trend with an overall level offset.
- **No published number for the residual after speed averaging.** Method 4 is physically sound and cheap, but if we use it we will be producing the first quantification we know of, not citing one.
- **No before-and-after numbers for converting an ordinary room** on a budget were found that met the evidence bar. Several web pages exist; none is a document with numbers we could quote.
- **No small-rotor facility was found in Africa, South America or Australia** despite dedicated searching. The correction to make here is that Africa is not absent from this guide: the Stellenbosch fan facility of §03 method 3 is South African, and it is the working rig whose correction method we propose to copy. What is missing is a small-rotor facility on those three continents, and South America appears only through general-purpose chambers. This is a limit of what our retrieval routes reach, not a claim that nothing exists.

The fetch list for the library proxy

PRIORITY	ITEM	WHY IT MATTERS	LINK
1	Zamponi, Gioli Torrione, Zůbek, Gallo, Zarri & Schram (2025), "Maneuvering drone noise investigation in a reverberant environment", *JASA* 157(6), 4468–4481	A rotor in a reverberant room, decontaminated with a tailored Green's function. Exactly method 3 applied to exactly our source. Marked open access; the publisher served a bot challenge.	doi.org/10.1121/10.0036900
2	Zamponi et al. (2024), "Aeroacoustic Assessment of the Noise Radiated by a Maneuvering Drone in an Indoor Test Zone", AIAA 2024-3274	The conference precursor to the above.	doi.org/10.2514/6.2024-3274
3	Cunefare, Badertscher & Wittstock (2006), "On the qualification of anechoic chambers: further issues related to signals and bandwidth", *JASA* 120(2), 820–829	The sequel to the tone-versus-noise paper that underpins §02. Treat the DOI as a lead, not verified.	doi.org/10.1121/1.2211467
4	Kopiev, Palchikovskiy et al. (2017), "Design and qualification of an anechoic facility in PNRPU", *Procedia Engineering* 176, 264–272	The peer-reviewed companion to the Perm chamber paper, with the room dimensions our table is missing. May be open access already.	Elsevier, Procedia Engineering vol. 176
5	Kloet, Watkins & Clothier (2017), "Acoustic signature measurement of small multi-rotor unmanned aircraft systems", *Int. J. Micro Air Vehicles* 9(2)	Would give Australia its first entry in the facility table.	doi.org/10.1177/1756829316681868
6	Singh, A. et al. (2025), "Characterization of an Anechoic Chamber for Conducting Aeroacoustic Experiments at IIT Kanpur"	Would give India a rotor-capable facility entry with hard numbers.	doi.org/10.1007/978-981-96-4082-9_11
7	Vanderkooy & Lipshitz (2008), "Can one perform quasi-anechoic measurements in normal rooms?", AES preprint 7525	The most-cited statement of §03 method 5's limits. AES library, paywalled.	aes2.org/publications/elibrary-page/?id=14677
8	ISO 13472-1 clause 7 (time-window length versus frequency range); ISO 3741 Annex E; ISO 5305 clauses 7.3.2–7.3.5 (microphone positions)	The actual numbers behind three methods we can currently only describe. Paywalled at PW; listed for completeness, expect nothing.	iso.org
9	GB 42590-2023 clause 4.14 "Noise" (full text, English or Chinese)	The only mandatory national drone-noise clause we found and could not read.	openstd.samr.gov.cn

Questions only you can answer

1. Room length, width and height, and the hub's height above the floor.
2. Foam: material, thickness, and how much of each surface it covers. The thickness alone decides, by the quarter-wave rule, the frequency above which it does anything.
3. What stands within 1.5 m of the microphones at −72° and −90°, and within 1.5 m of those at +36° and +54°. Those are the two named reflectors and they are physical objects.
4. How far below the propeller disc the thrust-stand beam sits, and on which side. The Italian result says a pylon offset to one side is enough to shift directivity by 4 dB.
5. Whether the university's electroacoustics department will lend us either an hour in their 300 m³ chamber or a laboratory sound source.

09 How this guide was built

Seven parallel research threads ran under a shared protocol: world facility inventory, qualification practice, alternatives to a chamber, the low-frequency wall, rig and inflow effects, small and low-cost chambers, and the standards landscape. Each thread was required to download a document and quote it with a page number or make no claim at all, to re-extract already-held papers with a plain text dump rather than trust earlier notes, and to log everything it could not obtain. Thirty-five new documents were downloaded and read; the evidence files are `papers/SoundVisualizer/chamber-landscape/EXTRACTS-*.md` and the retrieval logs sit beside them.

Three things in this guide are ours rather than the literature's, and are marked as such where they appear: the gating limit of 500 Hz and the 1.37 m requirement in §03, both computed from our own measured echo delays; the count of 29 and 55 cells outside the standard tolerance in §01, computed from our own position maps and offered as an analogy rather than a conformity result; and the Schroeder estimate of 190 to 270 Hz in §02, which depends on a room size we have not yet measured and is therefore a hypothesis.

Where this guide and the measurements disagree, the measurements govern. Where this guide and a quoted source disagree, the source governs and the error is ours. Quotations are verbatim from documents held in the private papers repository; page numbers are the source's own printed page where it has one. Facility numbers are as published by each facility and are not independently verified by us. This document is a working guide for our own decisions, not a review for publication, and it should be re-read against its sources before any of it is quoted elsewhere.